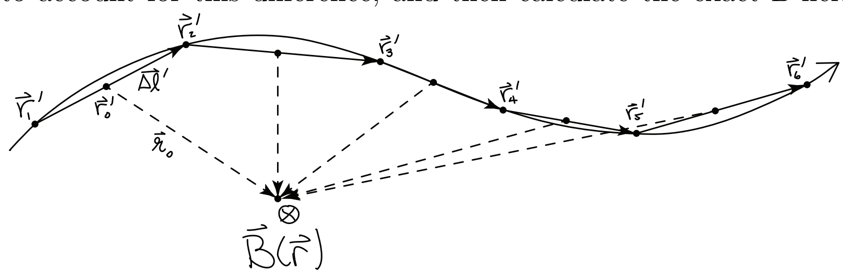
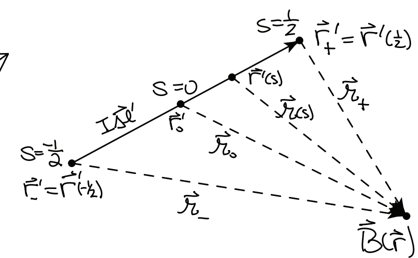
University of Kentucky, Physics 417G
Problem Set #1, Rev. A, due Monday, 2015-01-26

1. Thin wire—the Biot-Savart law can be integrated numerically by approximating a curved wire with a sequence of discrete straight current elements of small, but finite length:

$$\boldsymbol{B} \;=\; \frac{\mu_0}{4\pi}\oint' \frac{I\boldsymbol{d\ell}' \times \boldsymbol{\mathscr{r}}}{\mathscr{r}^3} \;\approx\; \sum_i \left(\Delta\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{I\Delta\boldsymbol{\ell} \times \boldsymbol{\mathscr{r}}_0}{\mathscr{r}_0^3}\right)_i, \tag{1}$$

where $\Delta\boldsymbol{\ell}$ is the displacement vector from the beginning to the end of each current segment, and $\boldsymbol{\mathscr{r}}_0$ is the displacement vector from the middle of each current segment $\boldsymbol{r}_0'$ to the field point $\boldsymbol{r}$. The approximation is that all of the current is concentrated at $\boldsymbol{r}_0'$ instead of spread out along the length of the segment from $\boldsymbol{r}_0' - \Delta\boldsymbol{\ell}/2$ to $\boldsymbol{r}_0' + \Delta\boldsymbol{\ell}/2$. In this problem we first calculate a correction term to account for this difference, and then calculate the exact B-field due to each straight segment.





a) To analytically integrate the Biot-Savart law along a single straight segment of the path, parametrize the segment $\boldsymbol{r}'(s)$ with the parameter s, ranging from $s = -\frac{1}{2}$ at the beginning to $s = +\frac{1}{2}$ at the end of the segment. The parametrization involves the constant vectors $\boldsymbol{r}_0'$ (the center of the segment) and $\Delta\boldsymbol{\ell}$ (displacement along the segment). Calculate the line element $\boldsymbol{dl} = \frac{dr'}{ds}ds$. Calculate $\boldsymbol{\mathscr{r}}$ as a function of $\boldsymbol{\mathscr{r}}_0$, $\Delta\boldsymbol{\ell}$, and s. Substitute these into the Biot-Savart formula and factor out the constants [essentially the approximation of Eq. 1] to obtain

$$\Delta\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu 0}{4\pi}\frac{I\Delta\boldsymbol{\ell}\times\boldsymbol{\mathscr{r}}_0}{\mathscr{r}_0^3}T(\alpha,\beta), \tag{2}$$

where the integral $T(\alpha,\beta)$ of s depends on the constants $\alpha = \boldsymbol{\mathscr{r}}_0 \cdot \Delta\boldsymbol{\ell}/\mathscr{r}_0^2$ and $\beta = \Delta\ell^2/\mathscr{r}_0^2$.

b) Approximate the integrand of $T(\alpha,\beta)$ to order s^2 and integrate to obtain the correction term $T(\alpha,\beta) \approx 1 + \frac{1}{8}(5\alpha^2 - \beta)$ for the case where all the current is at the center of the segment.

c) Calculate the exact integral $T(\alpha,\beta)$ and show that

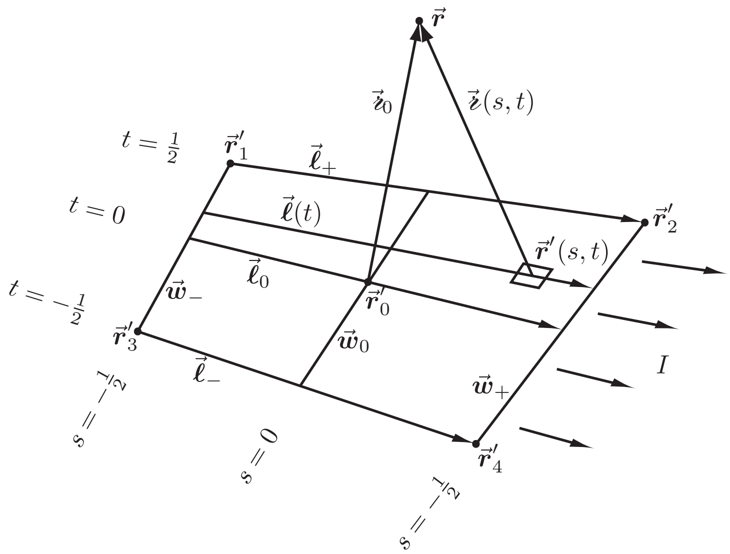
$$\Delta\boldsymbol{B} = \frac{\mu_0 I}{4\pi}\frac{\Delta\boldsymbol{\ell}\times\boldsymbol{\mathscr{r}}_0}{(\Delta\boldsymbol{\ell}\times\boldsymbol{\mathscr{r}}_0)^2}\,\Delta\boldsymbol{\ell}\cdot(\hat{\boldsymbol{\mathscr{r}}}_- - \hat{\boldsymbol{\mathscr{r}}}_+)\,, \tag{3}$$

where $\boldsymbol{\mathscr{r}}_\pm = \boldsymbol{r} - \boldsymbol{r}'(\pm\frac{1}{2})$ is the displacement vector from each end of the segment to the field point and $\hat{\boldsymbol{\mathscr{r}}}_\pm = \boldsymbol{\mathscr{r}}_\pm/\mathscr{r}_\pm$. Show that this agrees with Eq. (5.35) of Griffiths, and is equal to

$$\Delta\boldsymbol{B} = \frac{\mu_0 I}{4\pi}\frac{(\boldsymbol{\mathscr{r}}_-\times\boldsymbol{\mathscr{r}}_+)(\mathscr{r}_-+\mathscr{r}_+)}{\mathscr{r}_-\mathscr{r}_+\,(\mathscr{r}_-\mathscr{r}_+ + \boldsymbol{\mathscr{r}}_-\cdot\boldsymbol{\mathscr{r}}_+)} \tag{4}$$

[See "Compact expressions for the Biot-Savart fields of a filamentary segment", Eq. (9).]

2. Current sheet—surface currents can be approximated numerically by a tiling of quadrilaterals like the one shown below, with current I flowing parallel to the top and bottom edges, from left to right. Let the vector $\boldsymbol{\ell} = \boldsymbol{\ell}_+ = \boldsymbol{\ell}_-$ run along either the top or bottom edge, parallel to the current; $\boldsymbol{w} = \boldsymbol{w}_- = \boldsymbol{w}_+$ from bottom to top along the left or right edge; and $\boldsymbol{r}_0$ be the point at the center of the parallelogram as shown in the diagram.



a) Parametrize the surface of the parallelogram as $\boldsymbol{r}'(s,t)$, such that the top and bottom edges are at $t = +\frac{1}{2}$ and $-\frac{1}{2}$, respectively, and the left and right edges are at $s = +\frac{1}{2}$ and $-\frac{1}{2}$ respectively.

b) Write down the Biot-Savart integral for the magnetic field in terms of integration parameters s, t and constants $\boldsymbol{\mathscr{r}}_0 \equiv \boldsymbol{r} - \boldsymbol{r}'_0$, $\boldsymbol{\ell}$, and $\boldsymbol{w}$.

c) Expand the integrand in powers of s and t and calculate the integral up to second order.

Bonus: It is not possible to tile arbitrary surfaces with parallelograms—we need all four points on the quadrilateral to be arbitrary. To generalize this solution, let $\boldsymbol{\ell}_0$, $\boldsymbol{w}_0$ be the corresponding vectors through the center of the quadrilateral. We need one more vector $\boldsymbol{u}_0 = \boldsymbol{\ell}_+ - \boldsymbol{\ell}_- = \boldsymbol{w}_+ - \boldsymbol{w}_-$, where $\boldsymbol{\ell}_\pm$ run across the top and bottom, and $\boldsymbol{w}_\pm$ run along the right and left sides of the diagram. Generalize steps (a)-(c) to calculate $\boldsymbol{B}(\boldsymbol{r})$.

Also, Griffiths 3ed[4ed] Ch. 5, #3[3], 6[6], 7[7], 9[9], 11[11], 12[13].